



PUBLIC COURSEWORK DOCUMENTATION

LuntiAI

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THE PROBLEM

Three Barriers Holding Farmers Back

The Information Gap

Farmers rely on guesswork and tradition, causing **20-30% yield losses** annually from crop-soil mismatches.

The Accessibility Barrier

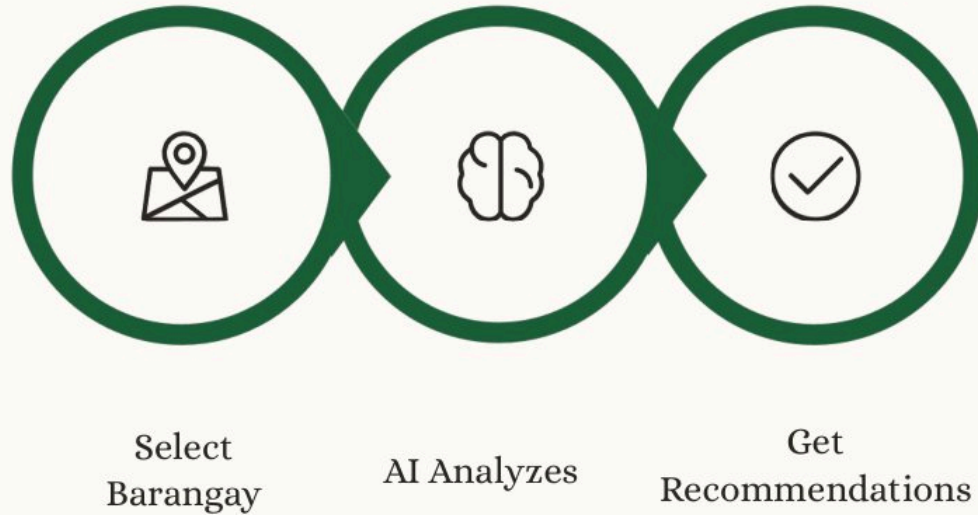
Lab soil testing costs **₱500-₱2,000** with a **2-4 week** wait — far beyond reach for farmers on 1-3 hectares.

Climate Vulnerability

El Niño/La Niña cycles make traditional calendars obsolete, yet no tool links **real-time weather** to crop selection.



How LuntiAI Works



LuntiAI eliminates manual data entry entirely — pre-mapping all 23 Tagum barangays and integrating live weather APIs so any farmer gets an accurate, localized recommendation in seconds.

AI Model & Data Architecture



Random Forest Classifier

85.62% accuracy across 12 Philippine crops — 200 estimators trained on 120,000+ domain-constrained synthetic samples.

Domain-Constrained Synthetic Data

Parameter ranges from **BSWM**, **PAGASA**, **PhilRice**, **DA Region XI**, and **PCA** ensure every training sample reflects real Philippine conditions.

Explainable AI (SHAP)

Transparent feature importance — farmers see exactly *why* a crop was recommended.

Grounded in Philippine Government Data

Every model parameter is sourced from authoritative Philippine institutions — ensuring recommendations are locally calibrated and scientifically valid.



Climate Data

PAGASA climatological normals + **OpenWeatherMap** live API for real-time temperature, humidity, and rainfall.



Crop Standards

PhilRice, **PCA**, **DA Mindanao**, **TADECO**, **PhilMaize**, and **PhilRootcrops** define optimal ranges for all 12 crops.



Soil Data

BSWM national soil profiles + **Tagum City Agriculture Office** barangay-level NPK, pH, and soil type mapping.



Financial Modeling

DA-CRA Region XI production costs + **DOLE Wage Order RB XI-24** (₱515/day) + **PSA** farmgate prices.

TARGET USER

Meet Mang Kiko — Our Primary User



Ricardo "Kiko" Dela Cruz, 48

Smallholder farmer · 2-hectare family farm ·
Pagsabangan, Tagum City · Android smartphone
user · ₱15,000–₱25,000/month income

"Mahirap mag-decide kung ano ang itanim
ngayon kasi iba na ang panahon."

Cannot afford

₱500+ lab soil tests

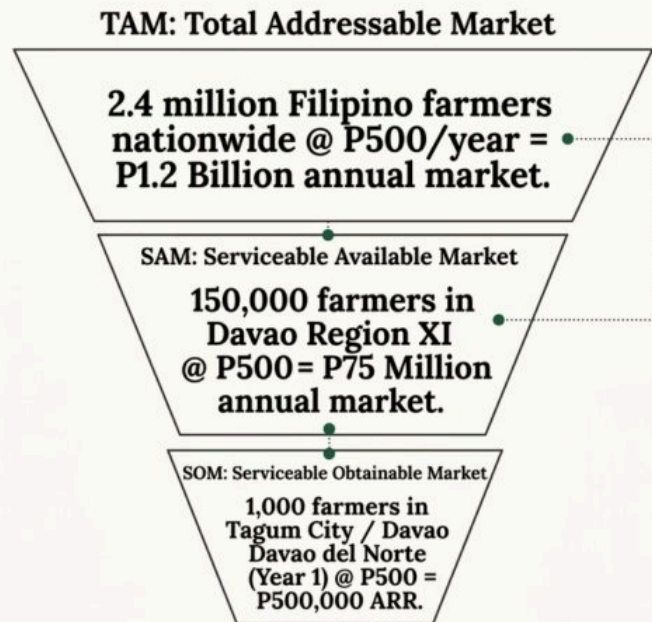
Needs

Tagalog/Bisaya interface

Wants

Accurate ROI using ₱515/day wage

TAM · SAM · SOM



Top-Down Market Sizing

Based on **RSBSA 2024** farmer registry data with a **₱500/year** premium subscription model.

- **TAM — ₱1.2B/yr:** 2.4M farmers nationwide
- **SAM — ₱75M/yr:** 150K Region XI farmers
- **SOM — ₱500K ARR:** 1,000 farmers, Year 1 beachhead

Five Revenue Streams

Freemium App

Basic crop recommendations, 3 queries/day — free, driving user acquisition

Premium Subscription

₱500/year — SHAP insights, DA-CRA ROI calculator, unlimited queries

B2B: Fertilizer Companies

₱50,000/quarter — anonymized barangay-level soil health data

B2G: City Agriculture

₱100,000/year — city-wide crop trend dashboards for LGUs

IoT Integration

₱2,500/sensor + ₱200/month — soil sensor hardware + data subscription

Why LuntiAI Wins

Feature	LuntiAI	Generic Crop Apps	Lab Soil Testing
Cost to Farmer	Free / ₱500/yr	Free	₱500–₱2,000/test
Speed	Instant	Instant	2–4 weeks
Localization	23 Tagum barangays	Generic / Global	N/A
Financial Accuracy	DA Region XI CRA	Hallucinated	N/A
Weather Integration	Live API	None	None
Explainability	SHAP AI insights	Black box	N/A

👍 LuntiAI is the only solution combining **barangay-level soil mapping, live weather, DA-CRA financial compliance, and explainable AI** in one mobile-first platform.



ROADMAP

The Path Forward

1

Q2 2026 — MVP Launch

Web app live: 12 crops, 23 barangays, multilingual UI (EN / TL / Bisaya)

2

Q3 2026 — Mobile App

Native Android app with offline-capable crop recommendations

3

Q4 2026 — IoT Pilot

NPK and OM soil sensor integration pilot in Tagum City

4

Q1 2027 — Regional Expansion

Scale to all Davao del Norte municipalities

Public roadmap version